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IT6203: IT Design Studio

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Milestone 2: Review Management Component

Review Management Component

Component Overview

The Review Management component allows renters to submit, view, edit, and delete reviews associated with rental properties. The component was developed using Angular for the frontend, Node.js and Express for backend functionality, and MongoDB for persistent data storage.

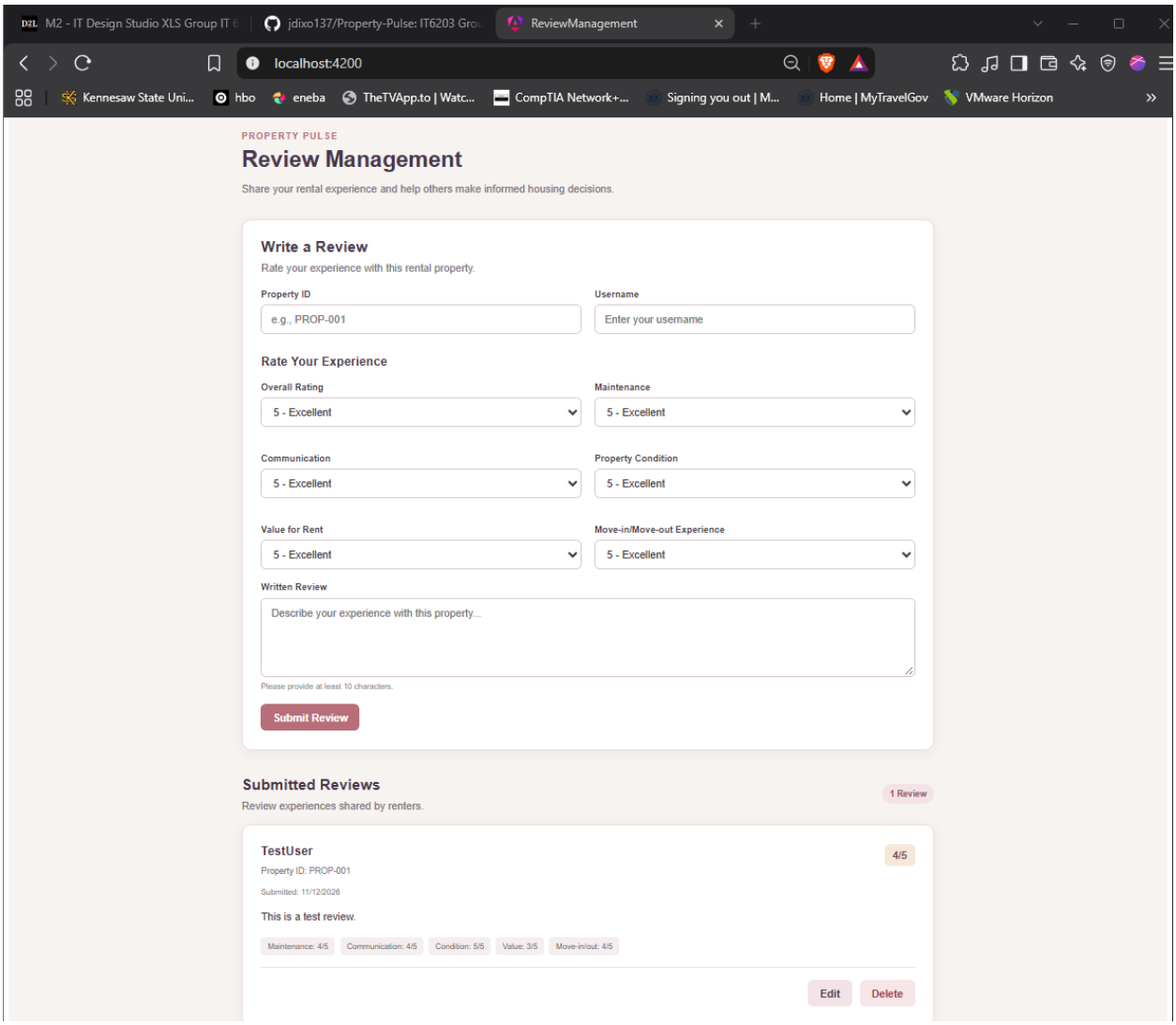
The application provides a reactive form that allows users to identify a property, enter a username, assign ratings across multiple categories, and submit a written review. The form includes validation to ensure required information is provided and written reviews contain at least ten characters.

The backend communicates with MongoDB through REST API endpoints, allowing review records to be stored and retrieved from the database. This implementation ensures that submitted reviews remain available after refreshing the application.

Screenshots and Descriptions

Read: Retrieving Reviews from MongoDB

When the application loads, the Angular component sends a GET request to the Express backend to retrieve existing reviews. The backend queries the MongoDB database and returns the stored review records. The submitted reviews are displayed with the renter's username, property ID, overall rating, individual category ratings, written review, and submission date.

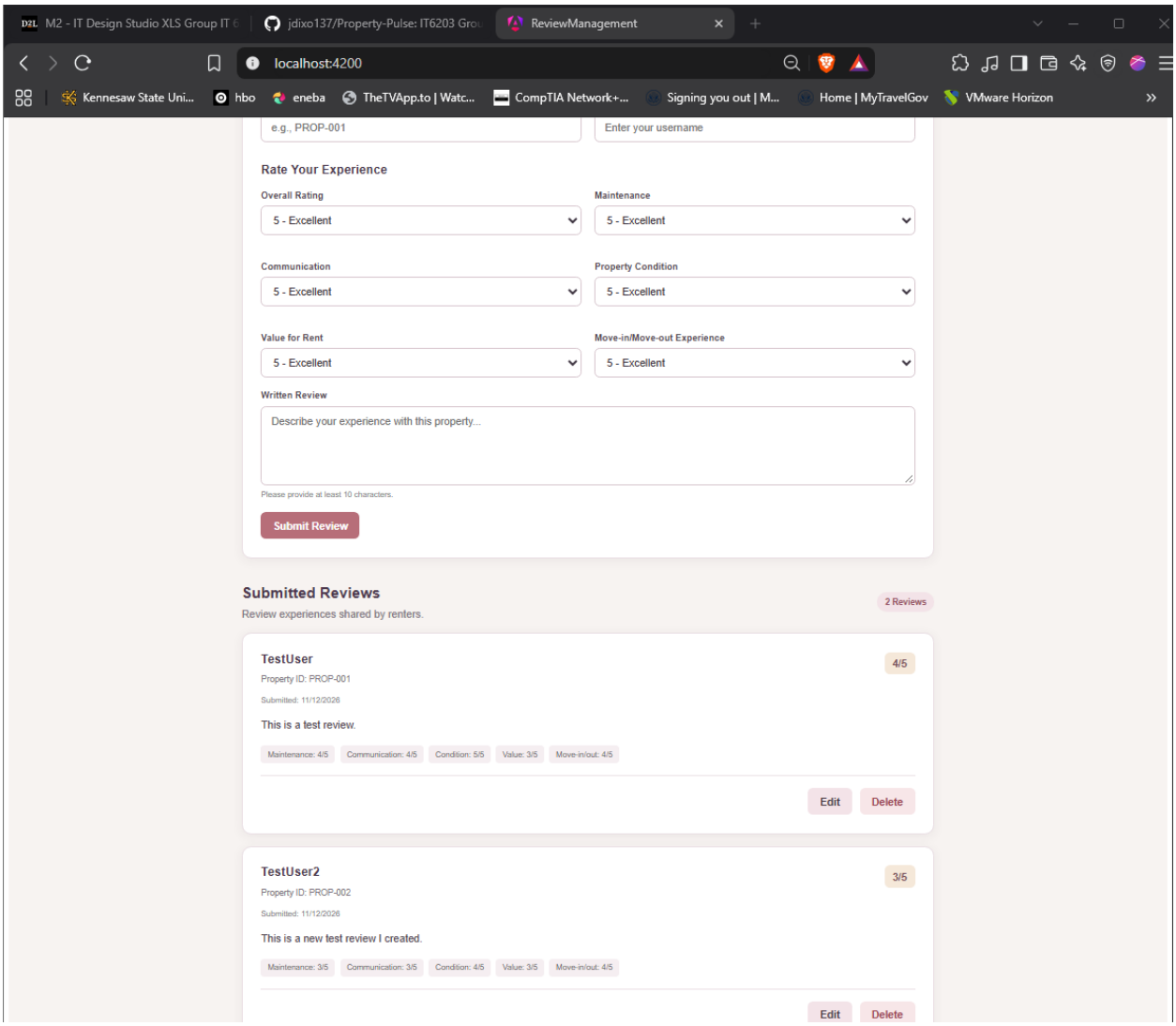


Create: Review Submission Form

The review submission form allows renters to provide property information, select ratings across six categories, and enter a written review. Reactive form validation ensures that required fields are completed and that written reviews meet the minimum character requirement.

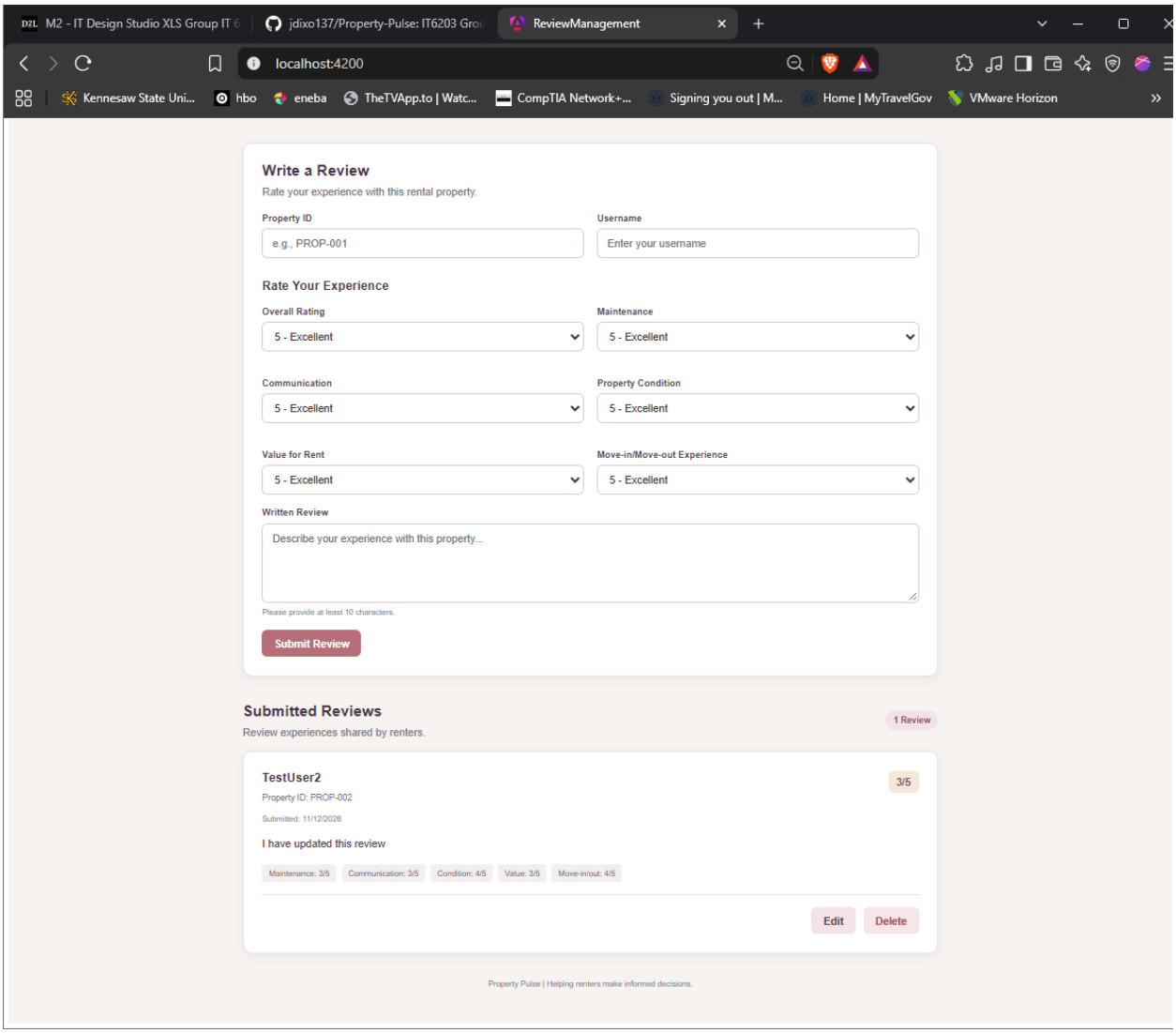
Create: Successfully Submitted Review

After completing the form, the user submits the review through the Angular application. The frontend sends a POST request to the Express API, which creates a new review document in MongoDB. The newly submitted review appears in the Submitted Reviews section, and the review counter updates accordingly.



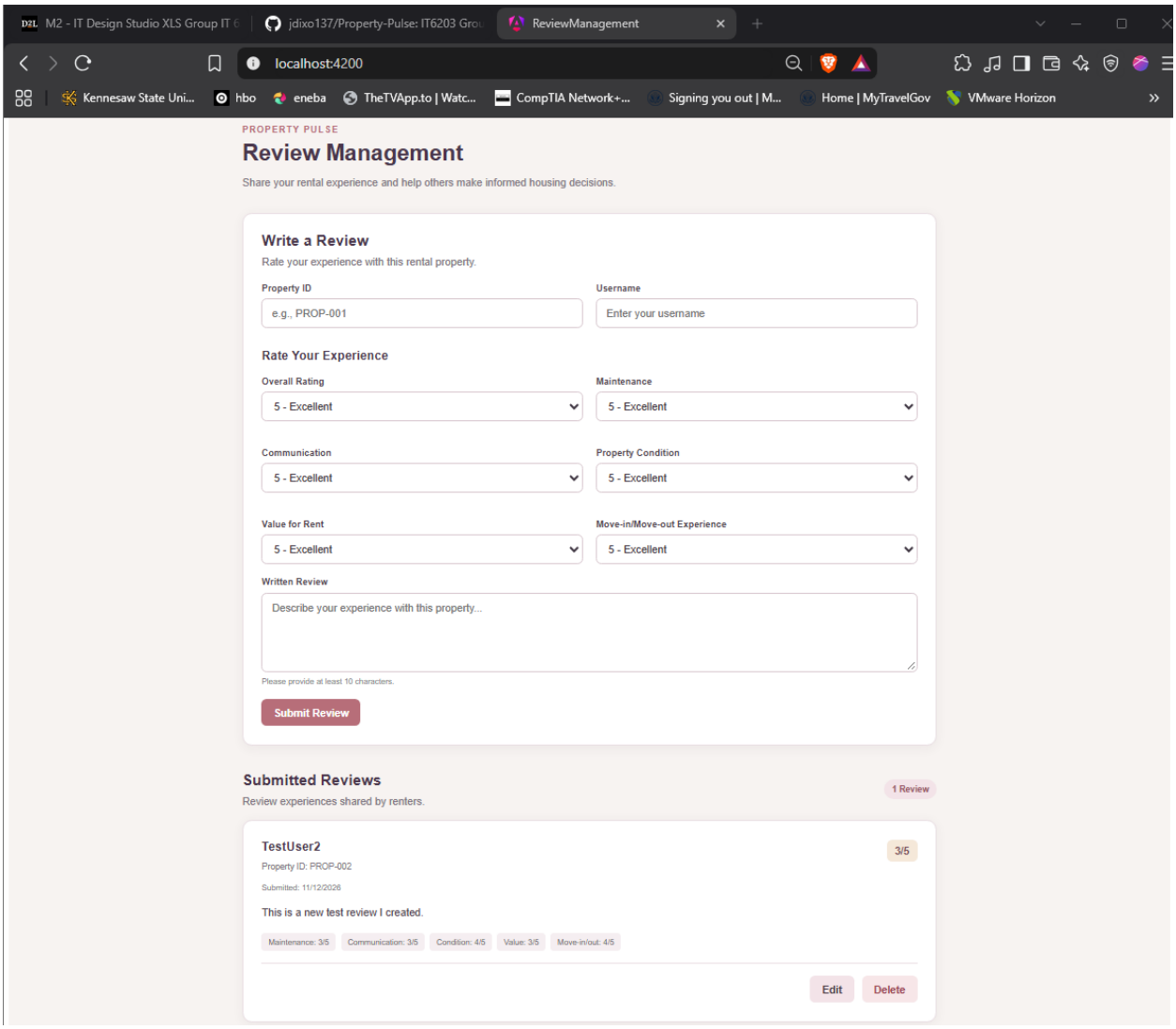
Update: Editing an Existing Review

Users can select the Edit button associated with an existing review. The application retrieves the selected review's information and populates the form with its current values. After making changes, the user selects Save Changes. The Angular application sends a PUT request to the backend, which updates the corresponding MongoDB document.



Delete: Removing an Existing Review

Users can remove an existing review by selecting the Delete button. The application requests confirmation before sending a DELETE request to the Express backend. The backend identifies the corresponding MongoDB document and removes it from the database. The application then refreshes the displayed review list to reflect the deletion.



Technologies and Implementation

The Review Management component uses the following technologies:

- **Angular:** Provides the user interface, reactive forms, and frontend functionality.
- **TypeScript:** Handles component logic, form validation, and communication with backend services.
- **Node.js:** Provides the backend runtime environment.
- **Express.js:** Implements REST API endpoints for review management.
- **MongoDB Community Server:** Stores review records persistently.
- **Mongoose:** Defines the review schema and manages database operations.

The backend implements four primary REST API operations:

Operation	HTTP Method	Endpoint
Create Review	POST	/api/reviews
Retrieve Reviews	GET	/api/reviews
Update Review	PUT	/api/reviews/:id
Delete Review	DELETE	/api/reviews/:id

Component Testing and Results

The Review Management component was tested to verify that all four CRUD operations function correctly with MongoDB.

Testing confirmed that newly submitted reviews were stored in the database and remained available after refreshing the application. Existing reviews could also be edited and deleted, with changes reflected in the database and frontend.

The component successfully demonstrates persistent data storage, backend communication, form validation, and functional review management.